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Battery pack

Abstract

A battery pack includes a pack case accommodating a plurality of battery modules, a bus bar coupled to two battery modules among the plurality of battery modules and interconnecting terminals provided in the two battery modules, and a bus bar cover accommodating the bus bar therein and coupled to the bus bar, wherein the bus bar cover is configured to surround the bus bar and entire upper and side surfaces.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
(1) This application claims benefit of priority to Korean Patent Application No. 10-2021-0039796 filed on Mar. 26, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.
BACKGROUND
1. Field
(2) Example embodiments of the present disclosure relate to a battery pack.
2. Description of Related Art
(3) Recently, a high-power battery pack using a non-aqueous electrolyte having high energy density

has been developed. Such a high-power battery pack may be implemented with large capacity by connecting a plurality of battery modules in series or in parallel such that the battery pack may be used to drive a motor of a device requiring high power, such as an electric vehicle, for example.

(4) In a general battery pack, a plurality of battery modules may be electrically connected through a bus bar, such that, when a flame occurs in the battery pack, insulation of the bus bar may be destroyed due to the flame and the bus bar may move, and accordingly, there may be a possibility that the bus bar may be in contact with a surrounding structure. In this case, an additional flame or explosion may occur due to a short circuit.

SUMMARY

(5) One embodiment of the present disclosure provides a battery pack which may prevent issues caused by a short circuit when a flame occurs in the battery pack.

(6) According to one embodiment of the present disclosure, a battery pack includes a pack case accommodating a plurality of battery modules, a bus bar coupled to two battery modules among the plurality of battery modules and interconnecting terminals of the two battery modules, and a bus bar cover accommodating the bus bar therein and coupled to the bus bar, wherein the bus bar cover is configured to surround the bus bar and entirely cover an upper surface and side surfaces of the bus bar.

(7) The bus bar cover may include an accommodation portion including an accommodation space for accommodating the bus bar, and an extension portion extending away from the accommodation portion to cover the side surfaces of the bus bar.

(8) The accommodation portion may include an upper surface portion opposing the upper surface of the bus bar, and a side surface portion extending from the upper surface portion and opposing the side surfaces of the bus bar.

(9) At least a portion of the side surface portion may be disposed between the bus bar and a case of the battery module.

(10) The extension portion may include a first extension portion disposed to oppose at least one of the side surfaces of the battery module on which the terminal is disposed, and a second extension portion disposed between the two battery modules.

(11) The pack case may include a reinforcing frame disposed between the battery modules, and the extension portion may include a frame insertion portion into which the reinforcing frame is inserted.

(12) The bus bar cover may include an accommodation portion including an accommodation space for accommodating the bus bar and covering entirely the upper surface and the side surfaces of the bus bar, and a coupling portion disposed below the bus bar and coupled to the accommodation portion.

(13) The battery pack may further include a module cover coupled to each of the battery modules, wherein the module cover accommodates a side surface of at least one of the battery modules on which the terminal is disposed.

(14) The module cover may have at least one slot into which the bus bar is inserted.

(15) The module cover may include mica.

(16) The module cover may be configured to cover an entirely of upper and side surfaces of the battery module formed in a rectangular parallelepiped shape.

(17) According to another embodiment of the present disclosure, a battery pack includes at least two battery modules, a bus bar electrically connecting the at least two battery modules to each other, and module covers coupled to the at least two battery modules, respectively, wherein each of the module covers accommodates one side surface of the battery module on which a terminal is disposed therein, and is coupled to the battery module.

(18) Each of the module covers may include at least one slot into which the bus bar is inserted.

(19) According to another embodiment of the present disclosure, a battery pack includes a pack case accommodating a plurality of battery modules, a bus bar coupled to two battery modules

among the plurality of battery modules and interconnecting terminals of the two battery modules, and a bus bar cover accommodating the bus bar therein and coupled to the bus bar, wherein the bus bar cover is configured to prevent flames or conductive particles in one battery module from spreading to an adjacent battery module.
(20) The bus bar cover may conform to a shape of the bus bar.
(21) The bus bar cover may include at least one of a refractory material or a ceramic material.
(22) The bus bar cover may include means for preventing shorting between the plurality of battery modules.
(23) The battery pack may further include a reinforcing frame disposed between the plurality of battery modules and a module cover coupled to at least one of the battery modules, wherein the module cover has a blocking portion disposed adjacent to the reinforcing frame to prevent the spreading of the flames or conducting particles to an adjacent battery module.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a perspective diagram illustrating a battery pack according to one embodiment of the present disclosure;
(3) FIG. **2** is an exploded perspective diagram illustrating a portion of the battery pack illustrated in FIG. **1**;
(4) FIG. **3** is a diagram illustrating a battery module illustrated in FIG. **1** of which a portion is omitted;
(5) FIG. **4** is a cross-sectional diagram taken along line I-I′ in FIG. **2**;
(6) FIG. **5** is an enlarged perspective diagram illustrating a bus bar cover illustrated in FIG. **2**;
(7) FIG. **6** is a perspective diagram illustrating the bus bar cover illustrated in FIG. **5**, viewed from the below;
(8) FIG. **7** is an enlarged perspective diagram illustrating a module cover illustrated in FIG. **2**;
(9) FIG. **8** is an exploded perspective diagram illustrating a portion of a battery pack according to another embodiment of the present disclosure;
(10) FIG. **9** is an exploded perspective diagram illustrating a bus bar cover illustrated in FIG. **8**;
(11) FIG. **10** is an exploded perspective diagram illustrating a portion of a battery pack according to still another embodiment of the present disclosure; and
(12) FIG. **11** is an exploded perspective diagram of FIG. **10**.
DETAILED DESCRIPTION
(13) It is to be understood that the terms or words used in this description and the following claims must not be construed to have meanings which are general or may be found in a dictionary. Therefore, considering that an inventor may most properly define the concepts of the terms or words to best explain his or her invention, the terms or words must be understood as having meanings or concepts that conform to the disclosed technical aspects the present disclosure. Also, since the embodiments set forth herein and the configurations illustrated in the drawings are nothing but a mere example(s) and are not representative of all technical aspects of the present disclosure, it is to be understood that various equivalents and modifications may replace the disclosed embodiments and configurations of the present application.
(14) In the drawings, same elements will be indicated by same reference numerals. Also, redundant descriptions and detailed descriptions of known functions and elements that may unnecessarily make the present disclosure obscure will be omitted. In the accompanying drawings, some

elements may be exaggerated, omitted or briefly illustrated, and the sizes of the elements do not necessarily reflect the actual sizes of these elements.

(15) For example, in embodiments disclosed, the terms “upper side,” “upper portion,” “lower side,” “lower portion,” and the like, are described with reference to the drawings, and when a direction of a corresponding element changes, the terms may be expressed differently.

(16) FIG. **1** is a perspective diagram illustrating a battery pack according to one embodiment of the present disclosure. FIG. **2** is an exploded perspective diagram illustrating a portion of the battery pack illustrated in FIG. **1**. FIG. **3** is a diagram illustrating a battery module illustrated in FIG. **1** of which a portion is omitted. FIG. **4** is a cross-sectional diagram taken along line I-I′ in FIG. **2**.

(17) Referring to FIGS. **1** to **4**, a battery pack **1** in one embodiment may include a pack case **2**, a plurality of battery modules **10**, a bus bar **20**, and a bus bar cover **50**.

(18) The pack case **2** may provide a space for accommodating the other components therein. In the drawings, for ease of description, the pack case **2** is illustrated in a form, shape, or configuration partially enclosing a lower portion and a side surface in the battery module **10**, but the pack case **2** may be provided to enclose the entirety of battery modules **10**. Accordingly, in one embodiment the pack case **2** may form an overall exterior of the battery pack **1**.

(19) Also, pack case **2** in the embodiment shown in FIG. **3** may include a reinforcing frame **3** in which the battery modules **10** are disposed therebetween.

(20) The reinforcing frame **3** may be provided to reinforce the rigidity of the pack case **2**. The reinforcing frame **3** may be disposed in a form, shape, or configuration dividing the space in which the battery modules **10** are disposed. Accordingly, in one embodiment, the reinforcing frame **3** may be disposed in between two battery modules **10** spaced apart from each other by a predetermined distance.

(21) The reinforcing frame **3** may be formed or otherwise configured in the form of a partition wall fastened to a bottom surface of the pack case **2**.

(22) The reinforcing frame **3** in the embodiment shown in FIG. **3** may include a first reinforcing frame **3***a* disposed to extend across both battery modules **10***a* and **10***c* and is arranged such that terminal surfaces TS (as shown in FIG. **4**) of a battery module **10** (on which terminal **15** is disposed or otherwise accommodated) may oppose each other in the battery module **10**. A second reinforcing frame **3***b* is disposed in between two battery modules **10***a* and **10***b*, as seen in FIG. **3**, to provide a space for accommodating battery modules **10***a* and **10***b*. More than one reinforcing frame **3***b* can be disposed side by side with each other. The terminal surfaces TS of battery module (s) **10** may be disposed facing the same direction as reinforcing frame **3***b* extends. Accordingly, the spaces in which the battery modules **10** are disposed may be partitioned by the reinforcing frames **3**.

(23) However, the present disclosure is not limited thereto, and only one of the first reinforcing frame **3***a* and the second reinforcing frame **3***b* may be provided if necessary.

(24) The reinforcing frame **3** may have a configuration in which the reinforcing frame **3** has a height lower than a height of the battery module **10**. For example, an upper end surface of the reinforcing frame **3** may be disposed in a position lower than the plane on which the terminal **15** is disposed.

(25) The reinforcing frame **3** may include a vent hole providing a gas flow path for discharging a gas (generated for example in thermal runaway) externally of the battery pack, such that high-pressure gas may be smoothly discharged during thermal runaway.

(26) In one embodiment of the present disclosure, the first reinforcing frame **3***a* and the second reinforcing frame **3***b* may be interconnected with each other. However, present invention is not limited thereto. In one embodiment of the present invention (such as shown in FIG. **4**), module cover **60** (coupled to at least one of the battery modules **10**) has a blocking portion (discussed in more detail later) extending downward from upper surface cover **62** and disposed adjacent to reinforcing frame to prevent spreading of flames to an adjacent battery module.

(27) Each of the battery modules **10** may have a hexahedral shape or configuration, and may be

arranged side by side in one direction. Each battery module **10** may include a battery such as a lithium secondary battery or a nickel-hydrogen secondary battery, which may be charged and discharged.

(28) As shown in FIG. **2**, each battery module **10** may be provided with terminal **15** on one side. The terminal **15** may be a conductive member exposed externally of the battery module **10** for electrical connection, and may be electrically connected to internal secondary batteries of the battery module **10**.

(29) The terminal **15** may be disposed parallel to a bottom surface of the pack case **2** on which the battery module **10** is seated. A lower surface of the terminal **15** may be supported by the case of the battery module **10**, and an upper surface of the terminal **15** may be opened externally of the battery module **10**.

(30) The terminal **15** may include a positive terminal and a negative terminal disposed spaced apart from each other.

(31) The bus bar **20** may be fastened to a plurality of the terminals **15** through a bus bar fastening member **90** (as shown in FIG. **2**). As the bus bar fastening member **90**, a member such as a bolt or a screw may be used, and the bus bar fastening member **90** may penetrate the bus bar **20** and the terminal **15** in order and may be fastened to the battery module **10**. Accordingly, the bus bar **20** may be pressed toward the terminal **15** by the bus bar fastening member **90** and may be physically/electrically connected to the terminal **15**.

(32) The bus bar **20** may interconnect (as shown in FIG. **2**) a terminal **15** of one battery module **10***a* to a terminal **15** of another battery module **10***b*. For example, the bus bar **20** may be in contact with a negative terminal **15** provided in the battery module **10***a* and a positive terminal **15** provided in another neighboring (or adjacent) battery module **10***b*, and may electrically/physically interconnect the terminals **15**.

(33) Accordingly, the plurality of battery modules **10** may be connected in series or in parallel by the bus bar **20**.

(34) As illustrated in FIG. **2**, the bus bar **20** according to this embodiment may have a conductive member in the shape or configuration of a substantially rectangular bar. Accordingly, when the bus bar **20** is coupled to the terminal **15**, a lower surface of the bus bar **20** may be disposed to oppose the terminal **15** and the upper surface may be disposed toward the outside of the battery module **10**.

(35) Since one bus bar **20** interconnects the terminals **15** of the two battery modules **10***b* and **10***a*, the bus bar **20** may be configured to have a length corresponding to the spacing distance between terminals **15**.

(36) The bus bar **20** may be formed of a material having flexibility. Also, the bus bar **20** may be formed in a form in which an insulating layer is coated on a surface of a conductive material (such as copper) to secure insulation from surrounding structures. However, the present invention is not limited thereto.

(37) FIG. **5** is an enlarged perspective diagram illustrating a bus bar cover illustrated in FIG. **2**. FIG. **6** is a perspective diagram illustrating the bus bar cover illustrated in FIG. **5**, viewed from the below.

(38) Referring to FIGS. **5** and **6** together, a bus bar cover **50** may accommodate the bus bar **20** therein and may be coupled to the bus bar **20**.

(39) The bus bar cover **50** may be provided to prevent a short circuit when the bus bar **20** is in contact with other structures and may prevent the occurrence of a flame in the battery module **10**. The bus bar cover **50** may also provide a function of preventing spread of flames or conductive particles to other adjacent battery modules **10**.

(40) The bus bar cover **50** in this embodiment may include an accommodation portion **51** and an extension portion **55**.

(41) The accommodation portion **51** may have the configuration of a cap and may have an accommodation space S which extend away from a top of the bus bar cover **50** and in the illustrated

embodiment in FIG. **6** opens downwardly. Therefore, in the process of assembling the battery pack **1**, the bus bar cover **50** may be disposed above the bus bar **20**, may move toward the bus bar **20** side and may accommodate the bus bar **20** in the accommodation space S, and may be coupled to or otherwise contact the bus bar **20**. Accordingly, the accommodation space S may be configured to have a size to fully accommodate the bus bar **20**. In this embodiment, the bus bar cover conforms to the shape of the bus bar **20**.

(42) The accommodation portion **51** may be configured to surround entire upper and side surfaces of the bus bar **20**. To this end, the accommodation portion **51** may include an upper surface portion **52** covering an upper surface of the bus bar **20** and a side surface portion **53** covering a side surface of the bus bar **20**.

(43) The upper surface portion **52** may be disposed to oppose the entire upper surface of the bus bar **20**. Also, the side surface portion **53** may extend from the upper surface portion **52** and may be disposed to oppose each of four side surfaces of the bus bar **20**. Accordingly, the side surface portion **53** may be formed to surround the entire side surface of the bus bar **20**.

(44) The side surface portion **53** of the bus bar cover **50** may include a second side surface **53***b* disposed between the bus bar **20** and the case of the battery module **10**. Accordingly, a space in which the second side surface **53***b* of the bus bar cover **50** may be disposed may be provided between the terminal **15** of the battery module **10** and the case of the battery module **10**.

(45) The extension portion **55** may be configured to extend in the downward direction from the accommodation portion **51**, and the extension portion **55** in the embodiment shown in FIG. **6** may include a first extension portion **55***a* and a second extension portion **55***b*.

(46) The first extension portion **55***a* may be disposed to oppose the terminal surface TS on which the terminal **15** is disposed among side surfaces of the battery module **10**. Accordingly, the first extension portion **55***a* may extend from the first side surface **53***a* disposed on an outermost side of the side surface portion **53**.

(47) In one embodiment of the present disclosure, the accommodation portion **51** may be simultaneously seated in between the two battery modules **10***a* and **10***b* interconnected by the bus bar **20**. Accordingly, the first extension portion **55***a* extending from the accommodation portion **51** may be disposed to oppose each of the two battery modules **10***a* and **10***b*. For example, the first extension portion **55***a* may be disposed to oppose a region of the terminal surface TS of the battery module **10**, disposed below the bus bar **20**.

(48) As described above, a second reinforcing frame **3***b* may be disposed between the battery modules **10**. Accordingly, the first extension portion **55***a* may include a frame insertion portion **56** into which the second reinforcing frame **3***b* is inserted.

(49) The frame insertion portion **56** may be formed as a groove in a configuration for cutting a relief in the first extension portion **55***a*. Also, to prevent a flame or conductive particles from moving into a gap formed between the frame insertion portion **56** and the second reinforcing frame **3***b*, the second reinforcing frame **3***b* may be coupled to the frame insertion portion **56** by being proximate to the frame insertion portion **56**.

(50) Accordingly, an overall shape of the frame insertion portion **56** may correspond to a cross-sectional surface of the second reinforcing frame **3***b* inserted into the frame insertion portion **56**. Also, a size or a shape (or a configuration) of the frame insertion portion **56** may be varied depending on a size or presence or absence of the second reinforcing frame **3***b*.

(51) The second extension portion **55***b* may be disposed between two battery modules **10***a* and **10***b* interconnected by a single bus bar **20**. Accordingly, the second extension portion **55***b* may extend from the second side surface **53***b* of the side surface portion **53**, disposed between the case of the battery module **10** and the bus bar **20**.

(52) Since the first side surface **53***a* and the second side surface **53***b* of the accommodation portion **51** are spaced apart from each other in a lateral direction by a predetermined distance and are arranged side by side, the first extension portion **55***a* and the second extension portion **55***b* may

also be spaced apart from each other and may be arranged side by side. However, an example embodiment thereof is not limited thereto.
(53) As the second reinforcing frame **3***b* is disposed between the battery modules **10***a* and **10***b*, the second extension portion **55***b* may have a configuration for blocking the space between the accommodation portion **51** and the second reinforcing frame **3***b*. Accordingly, a size of the second extension portion **55***b* may be varied depending on a size or presence or absence of the second reinforcing frame **3***b*.
(54) In one embodiment of the present disclosure, the extension portion **55** and the accommodation portion **51** may be configured to have the same thickness, but the present invention is not limited thereto. For example, the extension portion **55** and the accommodation portion **51** may be configured to have different thicknesses if desired. Also, various modifications of the second extension portion **55***b* may be made if desired such that the second extension portion **55***b* may extend along an upper surface of the second reinforcing frame **3***b*.
(55) The bus bar cover **50** may be formed of a flame retardant material or a fireproof insulating material which may block spread of a flame. In one example, the bus bar cover **50** may be formed of a material which does not melt or ignite below 1000° C., for example mica. Also, for increased heat resistance, the bus bar cover **50** may be formed from a plate material having a thickness of 1 mm or more. However, the present invention is not limited thereto.
(56) Through this configuration, the battery pack **1** in the one embodiment of the present disclosure may reduce a flame, a gas, and conductive particles generated in the battery pack **1** from affecting other battery modules **10**.
(57) Generally, the surrounding region of the terminal **15** may be formed of a resin material for electrical insulation in the battery module **10**, such that the surrounding region of the terminal **15** may be vulnerable to heat. Therefore, when thermal runaway or a flame occurs in the battery module **10**, a gas and conductive particles may be discharged externally of the battery module **10** through the terminal **15**.
(58) However, in battery pack **1** in one embodiment, since the bus bar **20** is accommodated in the accommodation space S of the bus bar cover **50**, most of flames or a gas and conductive particles discharged from the vicinity of the bus bar **20** may be prevented from flowing by the bus bar cover **50**.
(59) For example, in this bus bar cover **50**, flames or conductive particles generated in the battery pack **1** may be prevented from being emitted to the bus bar **20** side of an adjacent or neighboring battery modules **10** which are contact with the corresponding bus bar **20**.
(60) Also, since the first extension portion **55***a* of the bus bar cover **50** is extended from a front side of the bus bar **20** to a lower portion of the bus bar, inflow of flames or conductive particles into the bus bar cover **50** through the lower portion of the bus bar cover may be prevented.
(61) As described above, the battery pack **1** in one embodiment of the present disclosure may prevent the bus bar **20** from being directly exposed to flames or conductive particles, such that the removal (or destruction) of the insulating layer of the bus bar **20** caused by flames may be prevented, and accordingly, additional damages caused by a short circuit between the bus bar **20** and surrounding structures may be prevented.
(62) Also, the battery pack **1** in the example embodiment may further include a module cover **60**.
(63) FIG. **7** is an enlarged perspective diagram illustrating a module cover illustrated in FIG. **2**. Referring to FIG. **7**, the module cover **60** according to one embodiment of the present disclosure may be coupled to each battery module **10** by having a configuration for covering at least a portion of the battery module **10** and may prevent the spread of flames or conductive particles.
(64) The module cover **60** may accommodate a terminal surface TS of the battery module **10** having a rectangular parallelepiped shape, on which the terminal **15** is disposed, and may be coupled to the battery module **10**.
(65) For example, the module cover **60** may be formed in a shape surrounding the terminal surface

TS, and to this end, the module cover **60** may include a blocking portion **61** disposed to oppose the terminal surface TS of the battery module **10**, an upper surface cover **62** extending from the blocking portion **61** and disposed above the battery module **10**, and a side surface cover **63** extending from the upper surface cover **62** to oppose both side surfaces of the battery module **10**. Both side surfaces of the battery module **10** opposing the side surface cover **63** may refer to the side surfaces connected to the terminal surface TS of the battery module **10**.
(66) The upper surface cover **62** and the side surface cover **63** of the module cover **60** may be in contact with one battery module **10** and may be coupled to the battery module **10**. Also, when the module cover **60** is coupled to the battery module **10**, the blocking portion **61** of the module cover **60** may be disposed to be in contact with the terminal surface TS of the battery module **10** or may be disposed in a position adjacent to the terminal surface TS.
(67) As described above, since the battery modules **10** are interconnected by the bus bar **20**, when the module cover **60** is coupled to the battery module **10**, interference may occur between the module cover **60** and the bus bar **20**. Accordingly, the module cover **60** in the example embodiment may include at least one slot **65**.
(68) The slot **65** may have a configuration partially cutting the module cover **60**, and may be configured to have a width greater than a width of the bus bar **20** such that the bus bar **20** may be inserted there into.
(69) As described above, the bus bar cover **50** may be coupled to the bus bar **20**. Therefore, the slot **65** in one embodiment of the present disclosure may be configured to have a size for the bus bar cover **50** to be inserted there into.
(70) Referring to FIG. **2**, the bus bar **20** may connect two battery modules **10***a* and **10***b* arranged side by side. Accordingly, the slot **65** in the example embodiment may be formed on the side surface cover **63** disposed between the two battery modules **10** described above. Specifically, the slot **65** may be formed in a portion in which the blocking portion **61** and the side surface cover **63** are connected.
(71) In one embodiment of the present disclosure, the module cover **60** may include a high heat resistant material such as a refractory or ceramic material, for example mica. However, the present invention is not so limited to mica or a mica material, and the module cover **60** may be formed of a material the same as or similar to that of the bus bar cover **50** described above. For example, the module cover **60** may be formed of various materials as long as the module cover **60** may withstand flames or high temperatures.
(72) The module cover **60** in one embodiment of the present disclosure may be coupled to the battery module **10** together with the bus bar cover **50**. For example, the module cover **60** may be coupled to the battery module **10** after the bus bar cover **50** is coupled to the bus bar **20**. In this case, the bus bar cover **50** may be inserted into the slot **65** while being coupled to the bus bar **20**.
(73) As described above, when both the bus bar cover **50** and the module cover **60** are provided, the terminal **15** of the battery module **10** and the bus bar **20** may be disposed in a space formed by the module cover **60** and the bus bar cover **50**, such that, even when a flame is generated outside of the battery module **10**, an inflow of flames to the bus bar **20** (or the terminal **15**) may be prevented. Also, since the periphery of the terminal **15** is double protected by the bus bar cover **50** and the module cover **60**, a high reliability may be provided in terms of flame blocking.
(74) However, the present invention is not limited to the above-described battery pack. For example, the battery pack **1** may be configured to include only the module cover **60** without the bus bar cover **50**, or alternatively, the battery pack **1** may be configured to include only the bus bar cover **50** without the module cover **60**.
(75) Also, various modifications of the module cover **60** in the example embodiment may be made. In one embodiment, the module cover **60** may be configured to cover only one terminal surface TS, but the present invention is not limited thereto, and the module cover **60** may be configured to enclose the entire battery module **10** other than the bottom surface of the battery module **10**. For

example, the module cover **60** may be provided to oppose the entire upper and side surfaces of the battery module **10** and may be coupled to each battery module **10**. As described above, various modifications of the module cover **60** in these embodiments may be made as long as the module cover **60** may prevent an inflow of flames or conductive particles into the bus bar **20** side.
(76) FIG. **8** is an exploded perspective diagram illustrating a portion of a battery pack according to another embodiment. FIG. **9** is an exploded perspective diagram illustrating a bus bar cover illustrated in FIG. **8**.
(77) Referring FIGS. **8** and **9**, the battery pack in this embodiment may include a bus bar **20***a* (hereinafter, a second bus bar) for interconnecting two battery modules **10***a* and **10***c* arranged such that terminal surfaces of battery modules **10***a* and **10***c* oppose each other, in addition to the bus bar **20** (in FIG. **2**, hereinafter, a first bus bar).
(78) Accordingly, the bus bar cover **50** may include a bus bar cover **50** (hereinafter, a first bus bar cover) coupled to the first bus bar **20**, and a bus bar cover **50***a* coupled to the second bus bar **20***a*.
(79) Since the first bus bar **20** and the first bus bar cover **50** are of the same configuration as the bus bar **20** (in FIG. **2**) and the bus bar cover **50** (in FIG. **2**) of the above-described embodiment(s), the description(s) thereof will not be provided.
(80) The second bus bar **20***a* may electrically connect two battery modules **10***a* and **10***c* disposed above the first reinforcing frame **3***a* and having terminal surfaces opposing each other.
(81) The second bus bar cover **50***a* may include an accommodation portion **51** for accommodating the second bus bar **20***a* therein and coupled to the second bus bar **20***a*, and an extension portion **55** extending from the accommodation portion **51** in a downward direction.
(82) The extended portion **55** of the second bus bar cover **50***a* may be disposed in a space between the battery modules **10***a* and **10***c*. Since the first reinforcing frame **3***a* (in FIG. **3**) is disposed between the battery modules **10***a* and **10***c* arranged such that the terminal surfaces of the adjacent batteries oppose each other, the extension portion **55** of the second bus bar cover **50***a* may include a frame insertion portion **56** into which the first reinforcing frame **3***a* is inserted.
(83) The frame insertion portion **56** may be formed as a groove in the form of cutting the extension portion **55**. Also, to prevent flames or conductive particles from moving into a gap formed between the frame insertion portion **56** and the first reinforcing frame **3***a*, the first reinforcing frame **3***a* may be coupled to the frame insertion portion **56** by being proximate to the frame insertion portion **56**.
(84) Accordingly, an overall shape of the frame insertion portion **56** may correspond to a cross-sectional surface of the first reinforcing frame **3***a* inserted into the frame insertion portion **56**. Also, a size or a shape of the frame insertion portion **56** may be varied depending on a size or presence or absence of the first reinforcing frame **3***a*.
(85) When both the first bus bar **20** and the second bus bar **20***a* are coupled to a single battery module **10***a*, the module cover **60** coupled to the corresponding battery module **10***a* may include (as shown in FIG. **9**) a first slot **65***a* into which the bar **20** is inserted and a second slot **65***b* into which the second bus bar **20***a* is inserted. In this case, the first slot **65***a* may be formed on the side surface cover **63** and the second slot **65***b* may be formed on the blocking portion **61**.
(86) However, the present invention is not limited thereto, and the module cover **60** may include only one of the first slot **65***a* or the second slot **65***b* or may include both the first slot **65***a* and the second slot **65***b*, depending on the arrangement structure of the bus bar **20**.
(87) In the battery pack **1** in the configuration(s) described above, when a flame is generated in the battery module **10**, a short circuit may be prevented from occurring in an internal circuit of the battery pack **1** as the bus bar **20** moves.
(88) If a flame is generated in the battery module **10**, the case or other structure of the battery module **10** may be melted by heat, and the flame spread to the outside of the case of the battery module **10** may affect the other adjacent battery module. Also, as a structural collapse of the battery module **10** may occur in this process, the bus bar **20** may contact with the case of the battery module **10**, the pack case **2**, or other structures. In this case, a short circuit may occur in the internal

circuit of the battery pack **1**, which may cause additional fire or explosion.
(89) However, in the battery pack **1** according to one embodiment of the present disclosure, the bus bar **20** may be accommodated and disposed in an internal space of the bus bar cover **50** or the module cover **60**. Therefore, even when the bus bar **20** moves due to the collapse of the battery module **10** described above, the state in which the bus bar **20** is accommodated in the bus bar cover **50** or the module cover **60** may be maintained, such that the bus bar **20** may be prevented from being in contact with the case or surrounding structures. Accordingly, additional issues caused by a short circuit may be prevented.
(90) Also, since the battery pack **1** in one embodiment of the present disclosure may prevent spreading of flames by the module cover **60** or the bus bar cover **50**, melting of the insulating layer of the bus bar **20** and insulation of the bus bar **20** caused by flames directly in contact with the bus bar may be prevented, and accordingly, the insulation performance of the bus bar **20** may be continuously maintained even in the thermal runaway situation.
(91) FIG. **10** is an exploded perspective diagram illustrating a portion of a battery pack according to another example embodiment of the present disclosure. FIG. **11** is an exploded perspective diagram of FIG. **10**.
(92) Referring to FIGS. **10** and **11**, the bus bar cover **50***b* in this embodiment may be integrally coupled to the bus bar **20**.
(93) The bus bar cover **50***b* in this embodiment may include an accommodation portion **51***a* coupled to the upper surface of the bus bar **20** and a coupling portion **57** coupled to the lower surface of the bus bar **20**.
(94) The accommodation portion **51***a* may be coupled to the bus bar **20** by having a configuration covering an entire upper surface of the bus bar **20** and an entire side surface of the bus bar **20**. Also, the coupling portion **57** may be coupled to the bus bar **20** in a form of covering a portion of the bottom surface of the bus bar **20** other than a portion in which the bus bar **20** is in contact with the terminal **15**.
(95) Thus, when the accommodation portion **51***a* and the coupling portion **57** are coupled to the bus bar **20**, only a portion of the bus bar **20** in contact with the terminal **15** may be exposed externally of the bus bar cover **50***b*.
(96) Also, the bus bar cover **50***b* in this embodiment may further include a cover portion **58** to prevent the bus bar fastening member **90** from being exposed. The cover portion **58** may be coupled to the accommodation portion **51***a*, and may have a space in which the bus bar fastening member **90** is accommodated.
(97) The accommodation portion **51***a* may be provided with a coupling protrusion **59** protruding externally. The coupling protrusion **59** may be disposed along the periphery of the bus bar fastening member **90** disposed on the bus bar **20**, and the cover portion **58** may be inserted into the coupling protrusion **59** and may be coupled to the accommodation portion **51***a*.
(98) The bus bar cover **50***b* in this embodiment may be formed of a high temperature material such as a stainless steel, refractory or ceramic material. In this case, the bus bar cover **50***b* may be coupled to the bus bar **20** using an adhesive member. Also, an adhesive member may be interposed between the cover portion **58** and the accommodation portion **51***a*, but an example embodiment thereof is not limited thereto.
(99) Also, the bus bar cover **50***b* in this embodiment may be provided with a coating. For example, the accommodation portion **51***a* may be formed by applying a heat-resistant/insulating material, for example, Epoxy resin, to the upper surface of the bus bar **20**, and by applying a heat-resistant/insulating material to the lower surface of the bus bar **20**. In this case, the coupling protrusion **59** may not be provided.
(100) According to the aforementioned embodiment(s), when a flame is generated in the battery module, spread of the flame to the other battery module or addition issues caused by a short circuit may be reduced.

(101) While the example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A battery pack, comprising: a pack case accommodating a plurality of battery modules; a bus bar coupled to two battery modules among the plurality of battery modules and interconnecting terminals of the two battery modules; and a bus bar cover accommodating the bus bar therein and coupled to the bus bar, wherein the bus bar cover is formed as a unitary piece and configured to surround the bus bar and entirely cover an upper surface and side surfaces of the bus bar, wherein the pack case includes a first reinforcing frame disposed to oppose a terminal surface, which is at least one of side surfaces of the battery module on which a terminal is disposed, wherein the first reinforcing frame is disposed between the plurality of battery modules, wherein the bus bar cover includes an accommodation portion including an accommodation space for accommodating the bus bar and an extension portion extending away from the accommodation portion to cover the side surfaces of the bus bar, wherein the accommodation space includes an opening at a lower side thereof, and the bus bar cover is disposed over the bus bar such that the bus bar is accommodated inside the accommodation space through the opening, wherein the extension portion includes a first extension portion disposed to oppose the terminal surface, and wherein the first extension portion is disposed between the first reinforcing frame and the terminal surface.
2. The battery pack of claim 1, wherein the accommodation portion includes: an upper surface portion opposing the upper surface of the bus bar; and a side surface portion extending from the upper surface portion and opposing the side surfaces of the bus bar.
3. The battery pack of claim 2, wherein at least a portion of the side surface portion is disposed between the bus bar and a case of the battery module.
4. The battery pack of claim 1, wherein the extension portion includes: a second extension portion disposed between the two battery modules.
5. The battery pack of claim 1, wherein the pack case includes a second reinforcing frame disposed between the two battery modules, and wherein the first extension portion includes a frame insertion portion into which the second reinforcing frame is inserted.
6. The battery pack of claim 1, further comprising a second bus bar cover, wherein the second bus bar cover includes: an accommodation portion including an accommodation space for accommodating the bus bar and covering entirely the upper surface and the side surfaces of the bus bar; and a coupling portion disposed below the bus bar and coupled to the accommodation portion.
7. The battery pack of claim 1, further comprising: a module cover coupled to each of the battery modules, wherein the module cover accommodates the terminal surface.
8. The battery pack of claim 7, wherein the module cover has at least one slot into which the bus bar is inserted.
9. The battery pack of claim 7, wherein the module cover comprises mica.
10. The battery pack of claim 7, wherein the module cover is configured to cover an entirety of upper and side surfaces of the battery module formed in a rectangular parallelepiped shape.
11. A battery pack, comprising: at least two battery modules; a pack case accommodating the at least two battery modules; a bus bar electrically connecting the at least two battery modules to each other; module covers coupled to the at least two battery modules, respectively, wherein each of the module covers accommodates a terminal surface, which is at least one side surfaces of the battery module on which a terminal is disposed therein, and is coupled to the battery module, wherein the pack case includes a first reinforcing frame disposed to oppose a terminal surface, wherein a portion of the module cover is disposed between the first reinforcing frame and the terminal surface, further comprising a bus bar cover accommodating the bus bar therein and coupled to the

bus bar, wherein the bus bar cover includes an accommodation portion including an accommodation space for accommodating the bus bar and an extension portion extending away from the accommodation portion to cover the side surfaces of the bus bar, wherein the accommodation space includes an opening at a lower side thereof, and the bus bar cover is disposed over the bus bar such that the bus bar is accommodated inside the accommodation space through the opening, and wherein the pack case includes a second reinforcing frame disposed between the two battery modules, and wherein the extension portion includes a frame insertion portion into which the second reinforcing frame is inserted.

12. The battery pack of claim 11, wherein each of the module covers includes at least one slot into which the bus bar is inserted.

13. A battery pack, comprising: a pack case accommodating a plurality of battery modules; a bus bar coupled to two battery modules among the plurality of battery modules and interconnecting terminals of the two battery modules; and a bus bar cover accommodating the bus bar therein and coupled to the bus bar, wherein the bus bar cover is configured to prevent flames or conductive particles in one battery module from spreading to an adjacent battery module, wherein the bus bar cover includes an accommodation portion including an accommodation space for accommodating the bus bar and an extension portion extending away from the accommodation portion to cover the side surfaces of the bus bar, wherein the pack case includes a first reinforcing frame disposed to oppose a terminal surface, which is at least one of side surfaces of the battery module on which a terminal is disposed and a second reinforcing frame disposed between the two battery modules of the plurality of battery modules, wherein the extension portion of the bus bar cover is disposed between the first reinforcing frame and the terminal surface, wherein the extension portion includes a frame insertion portion into which the second reinforcing frame is inserted.

14. The battery pack of claim 13, wherein the bus bar cover conforms to a shape of the bus bar.

15. The battery pack of claim 13, wherein the bus bar cover comprises at least one of a refractory material, or a ceramic material.

16. The battery pack of claim 15, wherein the bus bar cover comprises mica.

17. The battery pack of claim 13, wherein the bus bar cover comprises means for preventing shorting between the plurality of battery modules.

18. The battery pack of claim 13, further comprising: a module cover coupled to at least one of the battery modules, wherein the module cover has a blocking portion disposed adjacent to the first reinforcing frame to prevent the spreading of the flames or conducting particles to an adjacent battery module.